QQ7 Phenol V2 (Legacy)

AQA 2020, 2021

IGC HK Exam

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0 4 Aspirin can be produced by reacting salicylic acid with ethanoic anhydride.
An incomplete method to determine the yield of aspirin is shown.

1. Add about 6 g of salicylic acid to a weighing boat.

2. Place the weighing boat on a 2 decimal place balance and record the mass.

3. Tip the salicylic acid into a 100 cm^3 conical flask.

4. ____

5. Add 10 cm^3 of ethanoic anhydride to the conical flask and swirl.

6. Add 5 drops of concentrated phosphoric acid.

7. Warm the flask for 20 minutes.

8. Add ice-cold water to the reaction mixture and place the flask in an ice bath.

9. Filter off the crude aspirin from the mixture and leave it to dry.

10. Weigh the crude aspirin and calculate the yield.

0 4 . 1 Describe the instruction that is missing from step **4** of the method.

Justify why this step is necessary.

[2 marks]

Instruction ____

Justification ____

____

____

0 4 . 2 Suggest a suitable piece of apparatus to measure out the ethanoic anhydride in step **5**.

[1 mark]

____

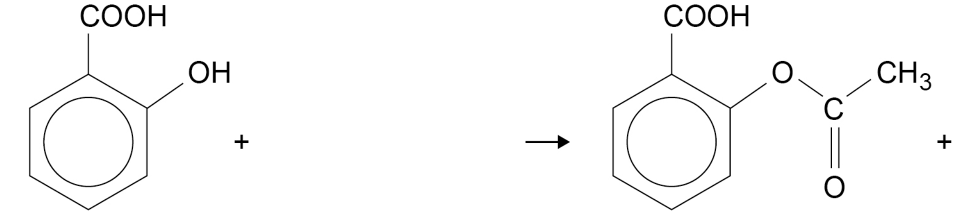
0 4 . 3 Identify a hazard of using concentrated phosphoric acid in step **6**.

[1 mark]

____

0 4 . 4 Complete the equation for the reaction of salicylic acid with ethanoic anhydride to produce aspirin.

[1 mark]



Salicylic acid Aspirin

0 4 . 5 A 6.01 g sample of salicylic acid (M_r = 138.0) is reacted with 10.5 cm^3 of ethanoic anhydride (M_r = 102.0).
In the reaction the yield of aspirin is 84.1%

The density of ethanoic anhydride is 1.08 g cm^{-3}

Show by calculation which reagent is in excess.

Calculate the mass, in g, of aspirin (M_r = 180.0) produced.

[5 marks]

Reagent in excess ______________________

Mass of aspirin ______________ g

Turn over ►

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0 4 . 6 Suggest **two** ways in which the melting point of the crude aspirin collected in step **9** would differ from the melting point of pure aspirin.

[2 marks]

Difference 1 ____

Difference 2 ____

0 4 . 7 The crude aspirin can be purified by recrystallisation using hot ethanol (boiling point = 78 °C) as the solvent.

Describe **two** important precautions when heating the mixture of ethanol and crude aspirin.

[2 marks]

Precaution 1 ____

Precaution 2 ____

0 4 . 8 The pure aspirin is filtered under reduced pressure.
A small amount of cold ethanol is then poured through the Buchner funnel.

Explain the purpose of adding a small amount of cold ethanol.

[1 mark]

0 4 . 9 A sample of the crude aspirin is kept to compare with the purified aspirin.

Describe **one** difference in appearance you would expect to see between these two solid samples.

[1 mark]

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Question	Answers	Additional comments/Guidelines	Mark
04.1	**M1** (Re)weigh the empty boat **M2** In order to calculate the (exact) mass of salicylic acid added to the reaction mixture		1 1
04.2	10 cm^3 measuring cylinder (if volume given – allow between 10 to 50 cm^3) Or a 10 cm^3 pipette Or burette / graduated pipette Or 10 cm^3 syringe		1
04.3	Corrosive	Allow skin burn / permanent eye damage Ignore irritant / toxic	1
04.4	LHS + $(CH_3CO)_2O$ RHS + CH_3COOH		1
04.5	**M1** Amount salicylic acid = $^{6.01}/_{138}$ = 4.36×10^{-2} mol **M2** Mass $(CH_3CO)_2O$ = 10.5 × 1.08 = 11.34 g **M3** Amount $(CH_3CO)_2O$ = $^{11.34}/_{102}$ = 1.11×10^{-1} mol **M4** $(CH_3CO)_2O$ is in excess **M5** Mass aspirin = M1 × 0.841 × 180 = 6.59 g	Allow conseq from wrong mole ratio in 04.4 Must show and state that ethanoic anhydride is in excess For M4/M5 ecf from M1/M3 Allow 2 sf or more.	1 1 1 1 1

04.6	**M1** Value lower **M2** Range of values	For M2 allow mpt not sharp or a larger range of melting points	1 1
04.7	**M1** (Ethanol is flammable so) use a water bath to heat / do not use a Bunsen burner **M2** Heat to temp below bp (so ethanol does not boil away)	Must give practical step, not just state hazard Allow use min vol solvent	1 1
04.8	To remove any soluble impurities	Allow To avoid aspirin dissolving (small amount cold solvent used) Allow To remove/(wash away) any ethanolic solution on the product.	1
04.9	Pure product will have (larger) crystals / needle-like crystals / lighter in colour	Allow whiter, less grey, more crystalline, less powdery, shinier, single colour Must be tied to pure product Allow opposite points tied to the crude product	1

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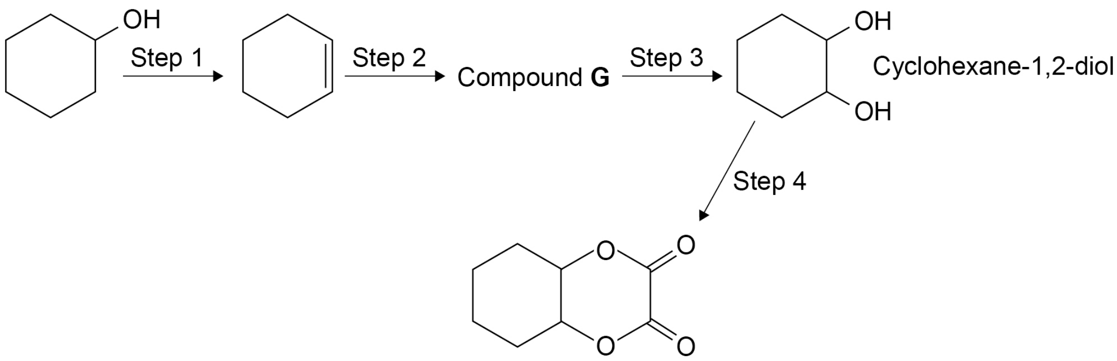
0 8 This question is about making a diester from cyclohexanol.



0 8 . 1 State the type of reaction in step **1**.

Give the name of the reagent needed for step **1**.

[2 marks]

Type of reaction ______________________________

Reagent ______________________________

0 8 . 2 State the reagents needed and give equations for step **2** and step **3**.

Show the structure of Compound **G** in your equations.

[4 marks]

Step 2 reagent ______________________________

Step 2 equation

____

Step 3 reagent ______________________________

Step 3 equation

____

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0 8 . 3 Cyclohexane-1,2-diol reacts with ethanedioyl dichloride.

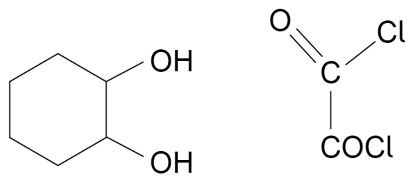
Give the name of the mechanism for this reaction.

Complete the mechanism to show the formation of **one** ester link in the first step of this reaction.

[5 marks]

Mechanism name ______________________________

Mechanism



0 8 . 4 Suggest why chemists usually aim to design production methods

- with fewer steps
- with a high percentage atom economy.

[2 marks]

Fewer steps ______________________________

____

____

High percentage atom economy ______________________________

____

____

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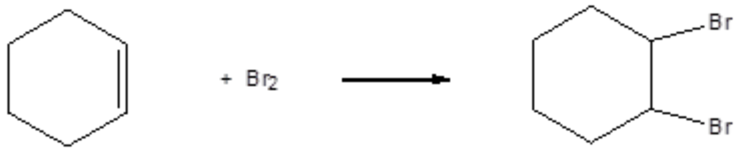
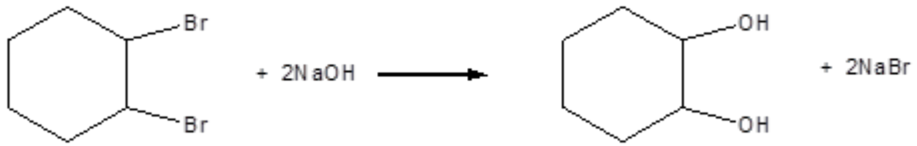
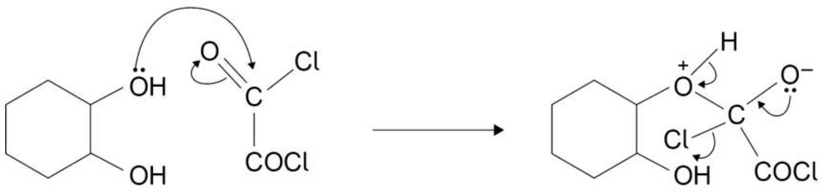
Question	Answers	Additional Comments/Guidelines	Mark
08.1	Dehydration	Allow (acid catalysed) Elimination	M1
	Conc H_2SO_4	Allow Conc H_3PO_4	M2

Question	Answers	Additional Comments/Guidelines	Mark
08.2	Br_2	Allow bromine (water) Allow Cl_2 or I_2 Allow O_2 if epoxide route used	M1
		allow conseq equation to H_2, H_2O, HBr, HCl. HI and H_2SO_4 An epoxide is a feasible alternative that could score here and consequentially M3 and M4	M2
	NaOH	**Or** KOH or other suitable strong alkali	M3
		Allow this equation with molecular formulae	M4

Question	Answers	Additional Comments/Guidelines	Mark
08.3	**M1** (nucleophilic)addition-elimination  **M2** curly arrow from lp on O to C **M3** curly arrow from double bond to O **M4** for structure of intermediate **M5** for 3 curly arrows	Note lone pair required for M5	1 M2 M3 M4 M5

Question	Answers	Additional Comments/Guidelines	Mark
08.4	Less energy used **OR** Better yield	**OR** reduces practical losses, simpler plant,	M1
	Less waste **OR** Less pollution	**OR** maximises the use of raw materials in the process into useful products, saves resources	M2